SCHEMATICITY APPEARS AT A FINITE STAGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1440-1449, following the statement at 1434-1438.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Opens descend to finite stages. lemma-descend-opens (same chapter, cited at ~1442-1443). Affine opens W_{j,i} in X_i with base change W_j.
- Affineness appears at a finite stage. lemma-limit-is-affine (same chapter, card L26293, cited at ~1444-1445). W_{j,i} may be taken affine.
- Affine covers make spaces schematic. Properties of Spaces, Section spaces-properties-section-schematic (cross-chapter, cited at ~1446-1448). A space with an affine open covering is a scheme.

Source reference. Original lemma 0827: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SQ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1370–1370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084R: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1250–1250. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03JG: focused-seeds/geometry/S9-spaces-limits-limit-is-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084R. If X is a scheme, then there exists an i such that X_i is a scheme.*

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a finite affine open covering $X = \bigcup W_j$. By Lemma 0827 we can find an $i \in I$ and open subspaces $W_{j,i} \subset X_i$ whose base change to X is $W_j \to X$. By Lemma 07SQ we may assume that each $W_{j,i}$ is an affine scheme. This means that X_i is a scheme (see for example Properties of Spaces, Section 03JG). $\square$